Chapter 1: Operating Systems

A process is a program in execution. Each process has its own address space.

The scheduler decides which process runs next on the CPU.

## Chapter 2: Memory

Paging divides memory into fixed-size pages. A page table maps virtual pages to frames.

A TLB caches recent translations to speed up address lookups.

## Chapter 3: Deadlocks

Deadlock needs mutual exclusion, hold and wait, no preemption and circular wait.

The Banker's algorithm avoids deadlock by checking for a safe state.